

MP5: Training Your Diffusion Model!

Setup environment

```
In [1]: # Import essential modules. Feel free to add whatever you need.
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
import torch # added
from torch import optim # added
from torch import nn
import torch.nn.functional as F
from torch.utils.data import DataLoader
from torchvision.datasets import MNIST
from torchvision.transforms import ToTensor
import torchvision.utils as vutils
import numpy as np
```

Visualization helper function

```
In [2]: def visualize_images_with_titles(images: torch.Tensor, column_names: list[str],
    """
    Visualize images as a grid and title the columns with the provided names.

    Args:
        images: (N, C, H, W) tensor of images, where N is (number of rows * num
        column_names: List of column names for the titles.
        font_size: Font size for the titles.

    Example usage:
    visualize_images_with_titles(torch.randn(16, 1, 32, 32), ['1', '2', '3', '4']
    """
    num_images, num_columns = images.shape[0], len(column_names)
    assert num_images % num_columns == 0, 'Number of images must be a multiple o

    num_rows = num_images // num_columns
    fig, axes = plt.subplots(num_rows, num_columns, figsize=(num_columns * 1, nu

    for i, ax in enumerate(axes.flat):
        img = images[i].permute(1, 2, 0).cpu().numpy()
        ax.imshow(img, cmap='gray')
        ax.axis('off')
        if i < num_columns:
            ax.set_title(column_names[i % num_columns], fontsize=font_size)

    plt.tight_layout()
    plt.show()

def image_add_noise(image: torch.Tensor, sigma: float):
    """
    Add noise to an image.

    Args:
```

```
image: (C, H, W) tensor of image
    Note: ToTensor() will naturally normalize the image to [0, 1]
sigma: Standard deviation of the Gaussian noise
```

```
Returns:
    Noisy image
"""
```

```
noisy_image = image + torch.randn_like(image) * sigma
# plt.imshow() will also naturally normalize the image to [0, 1]
# noisy_image = noisy_image.clamp(0., 1.)
return noisy_image
```

Part 1: Training a Single-step Denoising UNet

Implementing Simple and Composed Ops

```
In [3]: class Conv(nn.Module):
    def __init__(self, in_channels: int, out_channels: int):
        super().__init__()
        self.conv = nn.Conv2d(in_channels, out_channels, kernel_size=3, stride=1)
        self.bn = nn.BatchNorm2d(out_channels)
        self.gelu = nn.GELU()

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.conv(x)
        x = self.bn(x)
        x = self.gelu(x)
        return x

class DownConv(nn.Module):
    def __init__(self, in_channels: int, out_channels: int):
        super().__init__()
        self.conv = nn.Conv2d(in_channels, out_channels, kernel_size=3, stride=2)
        self.bn = nn.BatchNorm2d(out_channels)
        self.gelu = nn.GELU()

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.conv(x)
        x = self.bn(x)
        x = self.gelu(x)
        return x

class UpConv(nn.Module):
    def __init__(self, in_channels: int, out_channels: int):
        super().__init__()
        self.conv = nn.ConvTranspose2d(in_channels, out_channels, kernel_size=4,
        self.bn = nn.BatchNorm2d(out_channels)
        self.gelu = nn.GELU()

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.conv(x)
        x = self.bn(x)
```

```

        x = self.gelu(x)
        return x

class Flatten(nn.Module):
    def __init__(self):
        super().__init__()
        self.avg_pool = nn.AvgPool2d(kernel_size=7)
        self.gelu = nn.GELU()

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.avg_pool(x)
        x = self.gelu(x)
        return x

class Unflatten(nn.Module):
    def __init__(self, in_channels: int):
        super().__init__()
        self.conv = nn.ConvTranspose2d(in_channels, in_channels, kernel_size=7,
        self.bn = nn.BatchNorm2d(in_channels)
        self.gelu = nn.GELU()

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.conv(x)
        x = self.bn(x)
        x = self.gelu(x)
        return x

class ConvBlock(nn.Module):
    def __init__(self, in_channels: int, out_channels: int):
        super().__init__()
        self.conv1 = Conv(in_channels, out_channels)
        self.conv2 = Conv(out_channels, out_channels)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.conv1(x)
        x = self.conv2(x)
        return x

class DownBlock(nn.Module):
    def __init__(self, in_channels: int, out_channels: int):
        super().__init__()
        self.down_conv = DownConv(in_channels, out_channels)
        self.conv_block = ConvBlock(out_channels, out_channels)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.down_conv(x)
        x = self.conv_block(x)
        return x

class UpBlock(nn.Module):
    def __init__(self, in_channels: int, out_channels: int):
        super().__init__()
        self.up_conv = UpConv(in_channels, out_channels)
        self.conv_block = ConvBlock(out_channels, out_channels)

```

```
def forward(self, x: torch.Tensor) -> torch.Tensor:
    x = self.up_conv(x)
    x = self.conv_block(x)
    return x
```

Implementing Unconditional UNet

```
In [4]: class UnconditionalUNet(nn.Module):
    def __init__(
        self,
        in_channels: int,
        num_hiddens: int,
    ):
        super().__init__()
        self.conv_block1 = ConvBlock(in_channels, num_hiddens)
        self.conv_block2 = ConvBlock(2*num_hiddens, num_hiddens)

        self.down_block1 = DownBlock(num_hiddens, num_hiddens)
        self.down_block2 = DownBlock(num_hiddens, 2*num_hiddens)

        self.flatten = Flatten()

        self.unflatten = Unflatten(2*num_hiddens)

        self.up_block1 = UpBlock(4*num_hiddens, num_hiddens)
        self.up_block2 = UpBlock(2*num_hiddens, num_hiddens)

        self.conv = nn.Conv2d(in_channels=num_hiddens, out_channels=1, kernel_si

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        assert x.shape[-2:] == (28, 28), "Expect input shape to be (28, 28)." #
        # en_%d represents intermediate variables in encoder stage
        en1 = self.conv_block1(x)
        en2 = self.down_block1(en1)
        en3 = self.down_block2(en2)

        flat = self.flatten(en3)

        # de_%d represents intermediate variables in decoder stage
        # cat_%d represent concatenated intermediate variables from both encoder
        de1 = self.unflatten(flat)
        cat1 = torch.cat([de1, en3], dim=1)
        de2 = self.up_block1(cat1)
        cat2 = torch.cat([de2, en2], dim=1)
        de3 = self.up_block2(cat2)
        cat3 = torch.cat([de3, en1], dim=1)

        de4 = self.conv_block2(cat3)

        y = self.conv(de4)
        return y
```

Visualizing the noising process

```
In [5]: dataset = MNIST(root="data", download=True, transform=ToTensor(), train=True)
```



```

# TODO
target_digits = [7, 2, 1, 0, 4]
samples = []

for num in target_digits:
    idx = next(i for i, (_, label) in enumerate(dataset) if label == num)
    # img is tensor with shape (1, 28, 28)
    img, _ = dataset[idx]
    samples.append(img)

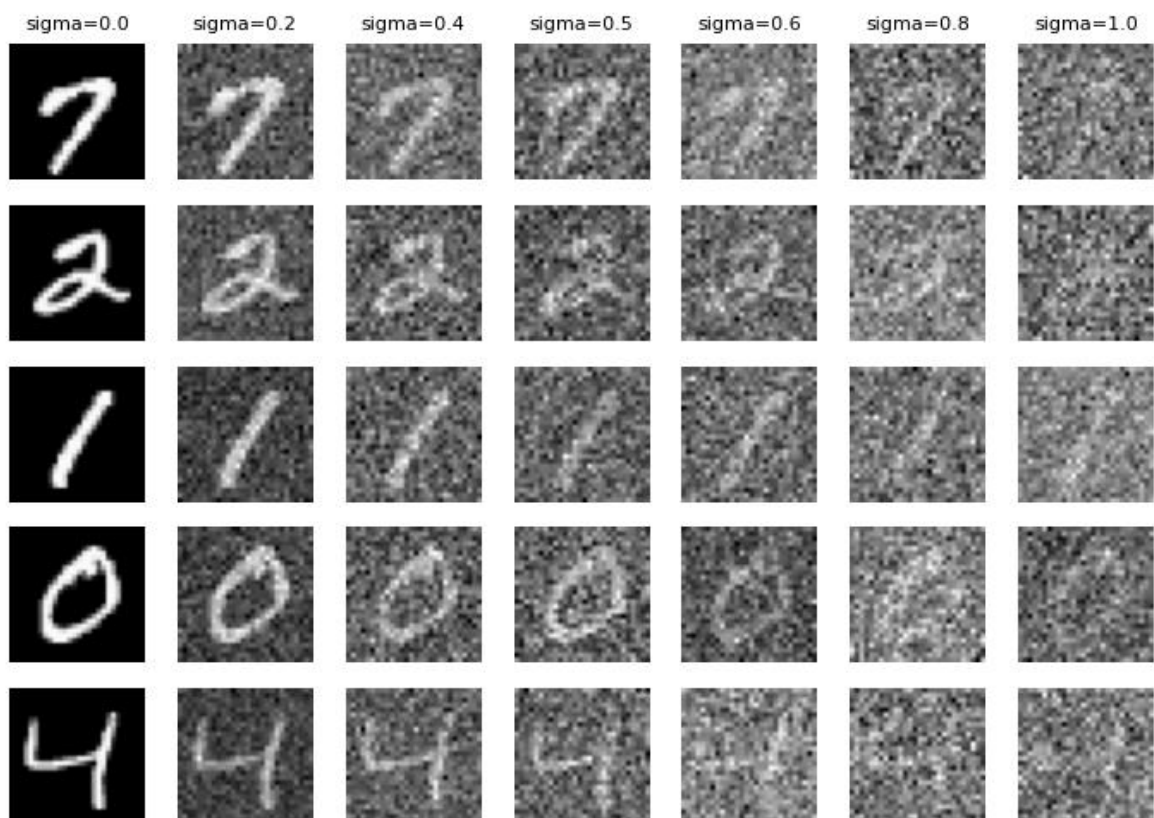
# Visualize difference noising processes
sigmas = [0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0]

noisy_samples = []
for sample in samples:
    for sigma in sigmas:
        noisy_sample = image_add_noise(sample, sigma)
        noisy_samples.append(noisy_sample)

noisy_samples_tensor = torch.stack(noisy_samples, dim=0)
# print(noisy_samples_tensor.shape)

visualize_images_with_titles(noisy_samples_tensor, [f'sigma={s:.1f}' for s in si

```



Training a Single-Step Unconditional UNet

- Plot the loss curve
- Sample results on the test set

```

In [6]: device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(device)

```

cuda

```
In [7]: train_dataset = MNIST(root='data', download=True, transform=ToTensor(), train=True)
train_dataloader = DataLoader(train_dataset, batch_size=256, shuffle=True)
test_dataset = MNIST(root='data', download=True, transform=ToTensor(), train=False)
test_dataloader = DataLoader(test_dataset, batch_size=3, shuffle=False)

# TODO
net = UnconditionalUNet(in_channels=1, num_hiddens=128).to(device)
optimizer = optim.Adam(net.parameters(), lr=1e-4)
criterion = nn.MSELoss()
```

```
In [8]: num_epochs = 5
# Record the losses for each batch/step
step_losses = []

# Training the UNet model
for epoch in range(5):
    net.train()
    for img, _ in train_dataloader:
        img = img.to(device)
        # Add noise to the images with sigma=0.5
        noisy_img = image_add_noise(img, sigma=0.5)

        # Forward pass
        pred = net(noisy_img)
        loss = criterion(pred, img)

        # Backward pass
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        step_losses.append(loss.item())

# Do visualization on epoch 1 and epoch 5
if epoch == 0 or epoch == 4:
    net.eval()
    img_test, _ = next(iter(test_dataloader))
    img_test = img_test.to(device)

    # Add noise to the test images
    noisy_img_test = image_add_noise(img_test, sigma=0.5)

    # Forward pass
    with torch.no_grad():
        denoised_img = net(noisy_img_test)

    # Move the images to CPU
    img_test = img_test.cpu()
    noisy_img_test = noisy_img_test.cpu()
    denoised_img = denoised_img.cpu()

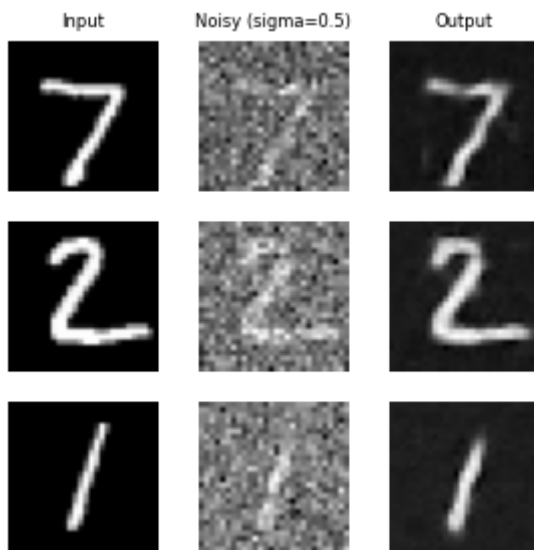
    # Visualize the origin images, noisy images and denoised images
    img_test_list = []
    for i in range(img_test.shape[0]):
        img_test_list.append(img_test[i])
        img_test_list.append(noisy_img_test[i])
        img_test_list.append(denoised_img[i])
    img_test_tensor = torch.stack(img_test_list, dim=0)
```

```

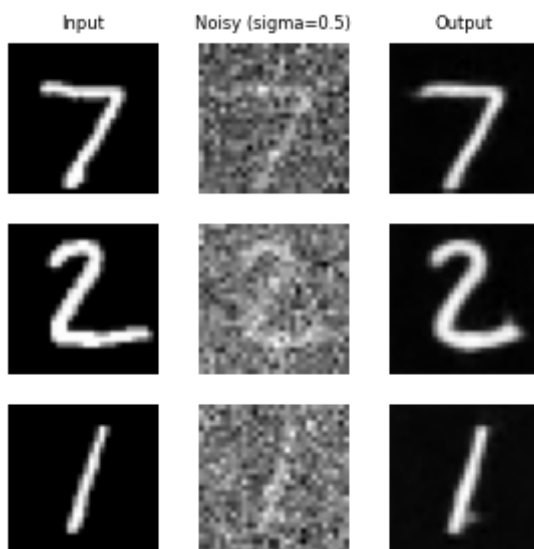
titles = ['Input', 'Noisy (sigma=0.5)', 'Output']
print(f"\n=== After epoch {epoch+1} ===")
visualize_images_with_titles(img_test_tensor, titles, 6)

```

=== After epoch 1 ===



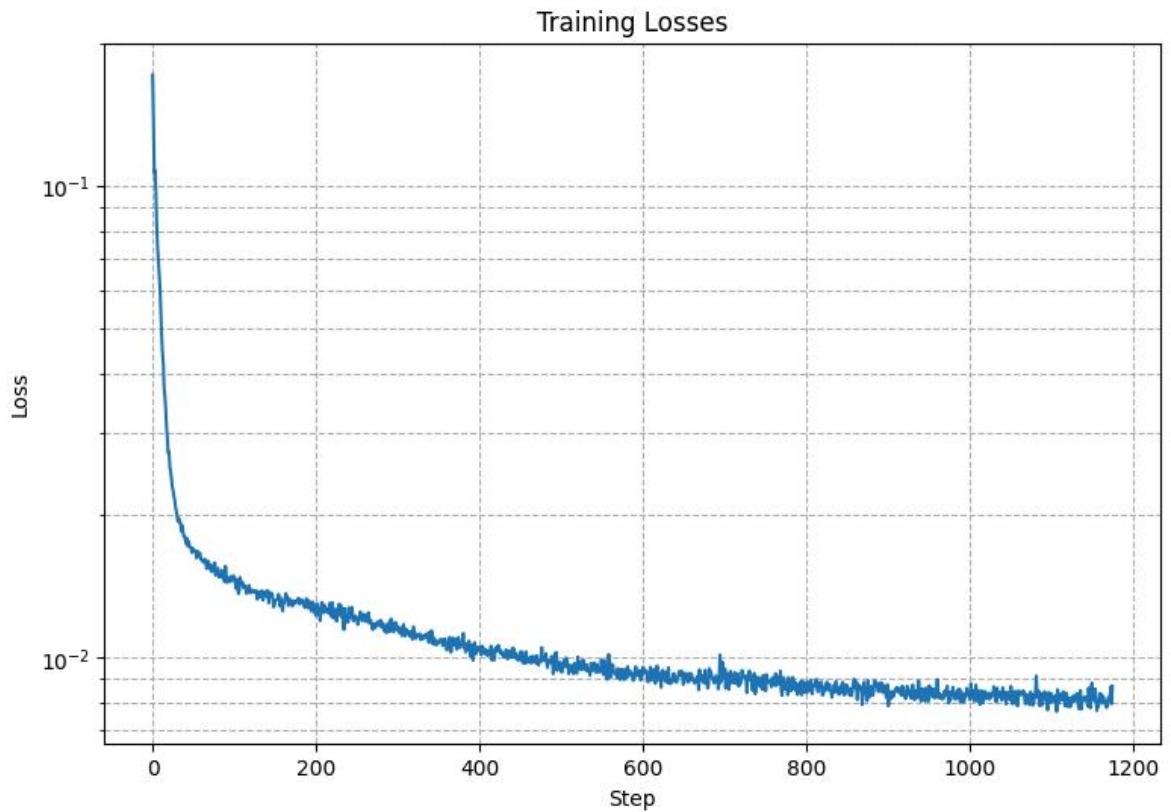
=== After epoch 5 ===



```

In [10]: # Plot training loss curves with respect to steps
plt.figure(figsize=(9, 6))
plt.semilogy(step_losses)
plt.xlabel('Step')
plt.ylabel('Loss')
plt.title('Training Losses')
plt.grid(True, which='both', ls='--')
plt.show()

```



Out-of-Distribution Testing

```
In [11]: # TODO
# Load the model
net.eval()
net.to(device)

# Load the data
idx = 0
img, _ = test_dataset[idx] # img: (1, 28, 28)
img = img.to(device)

# Visualize difference noising processes
sigmas = [0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0]

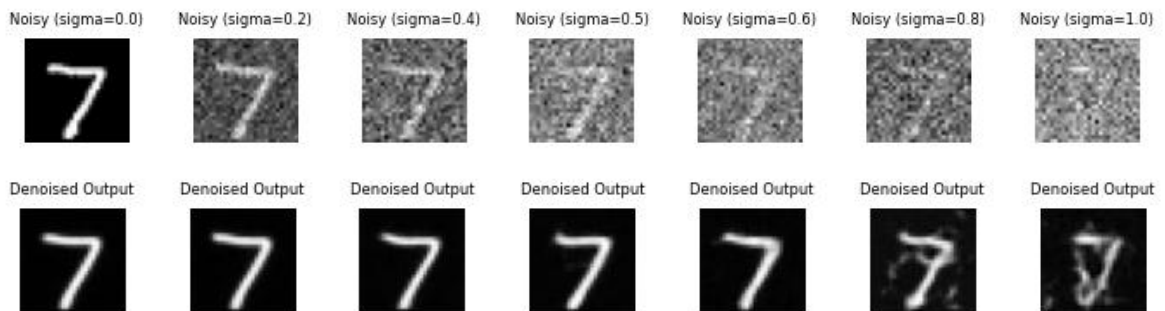
noisy_img_list = []
denoised_img_list = []
with torch.no_grad():
    for sigma in sigmas:
        # Add noise to the images
        noisy_img = image_add_noise(img, sigma)
        noisy_img = noisy_img.unsqueeze(0) # noisy_img: (1, 1, 28, 28)
        # Denoise the images
        denoised_img = net(noisy_img) # denoised_img: (1, 1, 28, 28)

        # Drop the first dimension, and save the images into lists
        noisy_img_list.append(noisy_img.cpu().squeeze(0))
        denoised_img_list.append(denoised_img.cpu().squeeze(0))

# Turn the List into tensors
noisy_img_tensor = torch.stack(noisy_img_list, dim=0)
denoised_img_tensor = torch.stack(denoised_img_list, dim=0)
```

```
# Visualize the noisy and denoised images
first_column_names = [f'Noisy (sigma={s:.1f})' for s in sigmas]
visualize_images_with_titles(noisy_img_tensor, first_column_names, 6)

second_column_names = ['Denoised Output' for _ in range(len(sigmas))]
visualize_images_with_titles(denoised_img_tensor, second_column_names, 6)
```



Part 2: Training a Diffusion Model

Helper function for generating .gif images

```
In [42]: import torch
import numpy as np
import torchvision.utils as vutils
from PIL import Image

def save_ddpm_gif(
    caches: torch.Tensor,
    gif_path: str,
    ncol: int = 4,
    nrow: int = 10,
    duration: int = 100
):
    """
    Save a DDPM denoising trajectory as a grayscale GIF, arranging each timestep
    """
    N, T1, C, H, W = caches.shape
    assert C == 1, "Only single-channel (grayscale) supported."

    frames = []
    for t in range(T1):
        # build the grid exactly as for static x_0 visualization
        big_grid = vutils.make_grid(caches[:, t], nrow=nrow, padding=0, normalize=True)

        # clamp pixel values to [0,1]
        big_grid = big_grid.clamp(0, 1) # (3, ncol*H, nrow*W)

        # convert the first channel to a uint8 numpy array in [0,255]
        arr = (big_grid[0].cpu().numpy() * 255).astype(np.uint8)

        # create a PIL grayscale image from the array
        img = Image.fromarray(arr, mode='L')
        frames.append(img)

    # prepend the cleanest frame, it becomes the static preview:
    clean_frame = frames[-1]
```

```
frames.insert(0, clean_frame)

# save the sequence of frames as an animated GIF
frames[0].save(
    gif_path,
    save_all=True,
    append_images=frames[1:],
    duration=duration,
    loop=0
)
```

Implementing a Time-conditioned UNet

```
In [43]: class FCBlock(nn.Module):
    def __init__(self, in_channels: int, out_channels: int):
        super().__init__()
        self.fc1 = nn.Linear(in_channels, out_channels)
        self.gelu = nn.GELU()
        self.fc2 = nn.Linear(out_channels, out_channels)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.fc1(x)
        x = self.gelu(x)
        x = self.fc2(x)
        return x

class TimeConditionalUNet(nn.Module):
    def __init__(self, in_channels: int, num_classes: int, num_hiddens: int):
        super().__init__()
        self.conv_block1 = ConvBlock(in_channels, num_hiddens)
        self.conv_block2 = ConvBlock(2*num_hiddens, num_hiddens)

        self.down_block1 = DownBlock(num_hiddens, num_hiddens)
        self.down_block2 = DownBlock(num_hiddens, 2*num_hiddens)

        self.flatten = Flatten()

        self.unflatten = Unflatten(2*num_hiddens)

        self.up_block1 = UpBlock(4*num_hiddens, num_hiddens)
        self.up_block2 = UpBlock(2*num_hiddens, num_hiddens)

        self.fcblock1 = FCBlock(1, 2*num_hiddens)
        self.fcblock2 = FCBlock(1, num_hiddens)

        self.conv = nn.Conv2d(in_channels=num_hiddens, out_channels=1, kernel_si

    def forward(self, x: torch.Tensor, t: torch.Tensor) -> torch.Tensor:
        """
        Args:
            x: (N, C, H, W) input tensor.
            t: (N,) normalized time tensor.

        Returns:
            (N, C, H, W) output tensor.
        """
        assert x.shape[-2:] == (28, 28), "Expect input shape to be (28, 28).\"
        # en_%d represents intermediate variables in encoder stage
```

```

en1 = self.conv_block1(x)
en2 = self.down_block1(en1)
en3 = self.down_block2(en2)

flat = self.flatten(en3)

# de_%d represents intermediate variables in decoder stage
# add_%d represents the time conditioning signal embedded in decoder stage
# cat_%d represent concatenated intermediate variables from both encoder and decoder
de1 = self.unflatten(flat)
# Transform t into form of (N, 1) because nn.Linear accepts 2D input
# If accept 1D input, it will broadcast to (1, N)
add1 = self.fcblock1(t.unsqueeze(-1)).reshape(t.shape[0], -1, 1, 1) # (N, 1, 1, 1)
de1 = de1 + add1
cat1 = torch.cat([de1, en3], dim=1)

de2 = self.up_block1(cat1)
add2 = self.fcblock2(t.unsqueeze(-1)).reshape(t.shape[0], -1, 1, 1) # (N, 1, 1, 1)
de2 = de2 + add2
cat2 = torch.cat([de2, en2], dim=1)

de3 = self.up_block2(cat2)
cat3 = torch.cat([de3, en1], dim=1)

de4 = self.conv_block2(cat3)

y = self.conv(de4)
return y

```

Implementing DDPM Forward and Inverse Process for Time-conditioned Denoising

```

In [44]: def ddpm_schedule(beta1: float, beta2: float, num_ts: int) -> dict:
    """Constants for DDPM training and sampling.

    Arguments:
        beta1: float, starting beta value.
        beta2: float, ending beta value.
        num_ts: int, number of timesteps.

    Returns:
        dict with keys:
            betas: linear schedule of betas from beta1 to beta2.
            alphas: 1 - betas.
            alpha_bars: cumulative product of alphas.
    """
    assert beta1 < beta2 < 1.0, "Expect beta1 < beta2 < 1.0."
    # Note that t is from 0 to num_ts(T)
    beta_tensor = torch.linspace(beta1, beta2, steps=num_ts+1) # beta_tensor: (num_ts+1,)
    alpha_tensor = 1 - beta_tensor # alpha_tensor: (num_ts+1,)
    # alpha_bar is the cumulative product of alphas
    alpha_bar_tensor = torch.cumprod(alpha_tensor, dim=0) # alpha_bar_tensor: (num_ts+1,)

    # Return the dict with keys
    result_dict = {}
    result_dict["betas"] = beta_tensor
    result_dict["alphas"] = alpha_tensor
    result_dict["alpha_bars"] = alpha_bar_tensor

```

```
return result_dict
```

```
In [45]: def ddpm_forward(
    unet: TimeConditionalUNet,
    ddpm_schedule: dict,
    x_0: torch.Tensor,
    num_ts: int,
) -> torch.Tensor:
    """Algorithm 1 of the DDPM paper.

    Args:
        unet: TimeConditionalUNet
        ddpm_schedule: dict
        x_0: (N, C, H, W) input tensor.
        num_ts: int, number of timesteps.
    Returns:
        (,) diffusion loss.
    """
    unet.train()
    # YOUR CODE HERE.
    N = x_0.shape[0]

    # Generate time steps randomly, and normalize to [0, 1]
    # Note in the algorithm, the random time steps are from 1 to num_ts(T)
    t = torch.randint(1, num_ts+1, (N,), device=x_0.device, dtype=torch.long)
    t_norm = t / num_ts

    # Generate gaussian noise
    eps = torch.randn_like(x_0)

    # Generate x_t with alpha_bar
    # The shape of t is (N,), so the shape of alpha_bar is (N, )
    alpha_bar = ddpm_schedule['alpha_bars'][t]
    alpha_bar = alpha_bar.reshape(N, 1, 1, 1) # alpha_bar: (N, 1, 1, 1)
    x_t = torch.sqrt(alpha_bar)*x_0 + torch.sqrt(1. - alpha_bar)*eps

    # Let UNet predict the noise
    pred_eps = unet(x_t, t_norm)

    # Calculate diffusion loss
    diffuse_loss = F.mse_loss(pred_eps, eps, reduction="mean")
    return diffuse_loss
```

```
In [46]: @torch.inference_mode()
def ddpm_sample(
    unet: TimeConditionalUNet,
    ddpm_schedule: dict,
    img_wh: tuple[int, int],
    num_ts: int,
    seed: int = 0,
) -> torch.Tensor:
    """Algorithm 2 of the DDPM paper with classifier-free guidance.

    Args:
        unet: TimeConditionalUNet
        ddpm_schedule: dict
        img_wh: (H, W) output image width and height.
        num_ts: int, number of timesteps.
```



```

seed: int, random seed.

Returns:
    (N, C, H, W) final sample.
    (N, T_animation, C, H, W) caches
    -> We have N=1 in this case.
"""
UNET.eval()
# YOUR CODE HERE.
H, W = img_wh
C = 1 # The number of image channels in MINST is 1
# Get device of UNet
UNET_device = next(UNET.parameters()).device

# Get tensors in ddpm_schedule
beta = ddpm_schedule['betas']
alpha = ddpm_schedule['alphas']
alpha_bar = ddpm_schedule['alpha_bars']

# Generate x_t(t=T) as pure noise
torch.manual_seed(seed)
x_t = torch.randn(C, H, W).unsqueeze(0) # x_T: (1, C, H, W)
x_t = x_t.to(UNET_device)
caches = [x_t.clone()]

# Stepwise backsampling from t=num_ts(T) to t=1
for t in range(num_ts, 0, -1):
    t_norm = torch.tensor([t], device=UNET_device, dtype=torch.float32) / num_ts
    # Predict noise
    pred_eps = UNET(x_t, t_norm)

    # Get beta_t/alpha_t/alpha_bar_t
    beta_t = beta[t]
    alpha_t = alpha[t]
    alpha_bar_t = alpha_bar[t]

    # Generate noise z
    if t > 1:
        z = torch.randn_like(x_t) # z: (1, C, H, W)
    else:
        z = torch.zeros_like(x_t) # z: (1, C, H, W)

    # According to the formula: x_{t-1} = 1/sqrt(alpha_t) [ x_t - (1-alpha_t)/sqrt(1-alpha_bar_t) * pred_eps ]
    coef_of_pred_eps = (1 - alpha_t) / torch.sqrt(1 - alpha_bar_t)
    x_t = (x_t - coef_of_pred_eps * pred_eps) / torch.sqrt(alpha_t) + torch.sqrt(1 - alpha_bar_t) * z

    # Add result into the cache List
    if t % 10 == 0:
        caches.append(x_t)

caches_tensor = torch.stack(caches, dim=0) # (T_animation, 1, C, H, W)
caches_tensor = caches_tensor.permute(1, 0, 2, 3, 4) # (1, T_animation, C, H, W)

return x_t, caches_tensor

```

```

In [47]: class DDPM(nn.Module):
def __init__(
    self,
    UNET: TimeConditionalUNET,
    betas: tuple[float, float] = (1e-4, 0.02),

```

```

num_ts: int = 300,
p_uncond: float = 0.1,
):
    super().__init__()
    self.unet = unet
    self.num_ts = num_ts
    self.p_uncond = p_uncond

    # Note we register alphas/alphaBars/betas as nn.ParameterDict, they will
    # (1) We can change to store them as buffer:
    # sched = ddpm_schedule(beta1, beta2, num_ts)
    # self.register_buffer('betas', sched['betas'])
    # self.register_buffer('alphas', sched['alphas'])
    # self.register_buffer('alphaBars', sched['alphaBars'])
    #
    # (2) We can calculate them locally in forward() and sample()...
    #
    # (3) Only include parameters in unet when setting Adam optimizer...
    self.ddpm_schedule = nn.ParameterDict(ddpm_schedule(betas[0], betas[1],

def forward(self, x: torch.Tensor) -> torch.Tensor:
    """
    Args:
        x: (N, C, H, W) input tensor.

    Returns:
        (,) diffusion loss.
    """
    return ddpm_forward(self.unet, self.ddpm_schedule, x, self.num_ts)

@torch.inference_mode()
def sample(
    self,
    img_wh: tuple[int, int],
    seed: int = 0,
):
    return ddpm_sample(self.unet, self.ddpm_schedule, img_wh, self.num_ts, s

```

Training the Time-conditioned UNet

- Plot the loss curve
- Sample results on the test set

```
In [48]: device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(device)
```

cuda

```
In [49]: train_dataset = MNIST(root='data', download=True, transform=ToTensor(), train=True)
train_dataloader = DataLoader(dataset, batch_size=128, shuffle=True)
test_dataset = MNIST(root='data', download=True, transform=ToTensor(), train=False)
test_dataloader = DataLoader(dataset, batch_size=40, shuffle=True)

# TODO
net = TimeConditionalUNet(in_channels=1, num_classes=0, num_hiddens=64)
model = DDPM(net, betas=(0.0001, 0.02), num_ts=300, p_uncond=0.0).to(device)

# Determine Loss function, scheduler and optimizer
```

```

from torch.optim.lr_scheduler import ExponentialLR

num_epochs = 20
learning_rate = 1e-3
gamma = 0.1 ** (1/num_epochs)

optimizer = optim.Adam(model.unet.parameters(), lr=learning_rate)
scheduler = ExponentialLR(optimizer, gamma=gamma)
# criterion = nn.MSELoss()

```

```

In [50]: # Record the losses for each batch/step
step_losses=[]
checkpoints = {0, 4, 9, 14, 19}

# Train the diffusion model
for epoch in range(0, num_epochs):
    model.train()

    for x_0, _ in train_dataloader:
        x_0 = x_0.to(device)

        # Forward pass
        # Note that DDPM forward() returns the diffusion loss
        loss = model(x_0)

        # Backward pass
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        step_losses.append(loss.item())

    # Update the Learning rate
    scheduler.step()

    # Calculate the average loss over the current epoch
    avg_loss = sum(step_losses[-len(train_dataloader):]) / len(train_dataloader)
    print(f"Epoch {epoch+1:2d} | avg train loss: {avg_loss:.4e}")

    # Visualize on certain epochs
    if epoch in checkpoints:
        model.eval()
        n_rows, n_cols = 4, 10
        num_samples = n_rows * n_cols
        samples = []
        caches = []
        for seed in range(num_samples):
            img, cache = model.sample((28,28), seed=seed) # img: (1, 1, 28, 28)
            samples.append(img.cpu().squeeze(0)) # (1, 28, 28)
            caches.append(cache.cpu().squeeze(0)) # (301, 1, 28, 28)
        grid = torch.stack(samples, dim=0) # (40, 1, 28, 28)
        cache_grid = torch.stack(caches, dim=0) # (40, 301, 1, 28, 28)

        save_ddpm_gif(cache_grid, f"gif/Time_UNet_epoch_{epoch+1}.gif", n_rows,

        big_grid = vutils.make_grid(grid, nrow=n_cols, padding=0, normalize=False)
        # print(big_grid.shape)

        plt.figure(figsize=(10, 4))
        plt.imshow(big_grid[0].cpu(), cmap='gray', vmin=0, vmax=1)

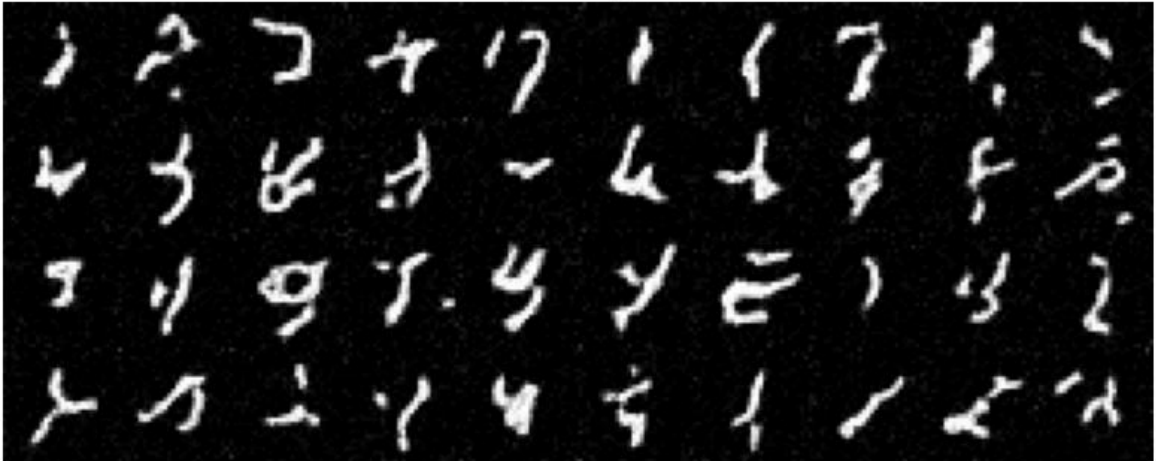
```

```
plt.axis('off')
plt.title(f"Samples at Epoch {epoch+1}")
plt.show()

# fig, axes = plt.subplots(n_rows, n_cols, figsize=(n_cols, n_rows))
# for i, ax in enumerate(axes.flat):
#     ax.imshow(grid[i], cmap='gray')
#     ax.axis('off')
# plt.suptitle(f"Samples at Epoch {epoch+1}")
# plt.tight_layout()
# plt.show()
```

Epoch 1 | avg train loss: 7.1982e-02

Samples at Epoch 1



Epoch 2 | avg train loss: 4.3873e-02

Epoch 3 | avg train loss: 3.7977e-02

Epoch 4 | avg train loss: 3.5299e-02

Epoch 5 | avg train loss: 3.3524e-02

Samples at Epoch 5



Epoch 6 | avg train loss: 3.2320e-02

Epoch 7 | avg train loss: 3.0744e-02

Epoch 8 | avg train loss: 3.0175e-02

Epoch 9 | avg train loss: 2.9519e-02

Epoch 10 | avg train loss: 2.9025e-02

Samples at Epoch 10



Epoch 11 | avg train loss: 2.8882e-02
 Epoch 12 | avg train loss: 2.8119e-02
 Epoch 13 | avg train loss: 2.7824e-02
 Epoch 14 | avg train loss: 2.7632e-02
 Epoch 15 | avg train loss: 2.7303e-02

Samples at Epoch 15



Epoch 16 | avg train loss: 2.7440e-02
 Epoch 17 | avg train loss: 2.6836e-02
 Epoch 18 | avg train loss: 2.6685e-02
 Epoch 19 | avg train loss: 2.6555e-02
 Epoch 20 | avg train loss: 2.6332e-02

Samples at Epoch 20



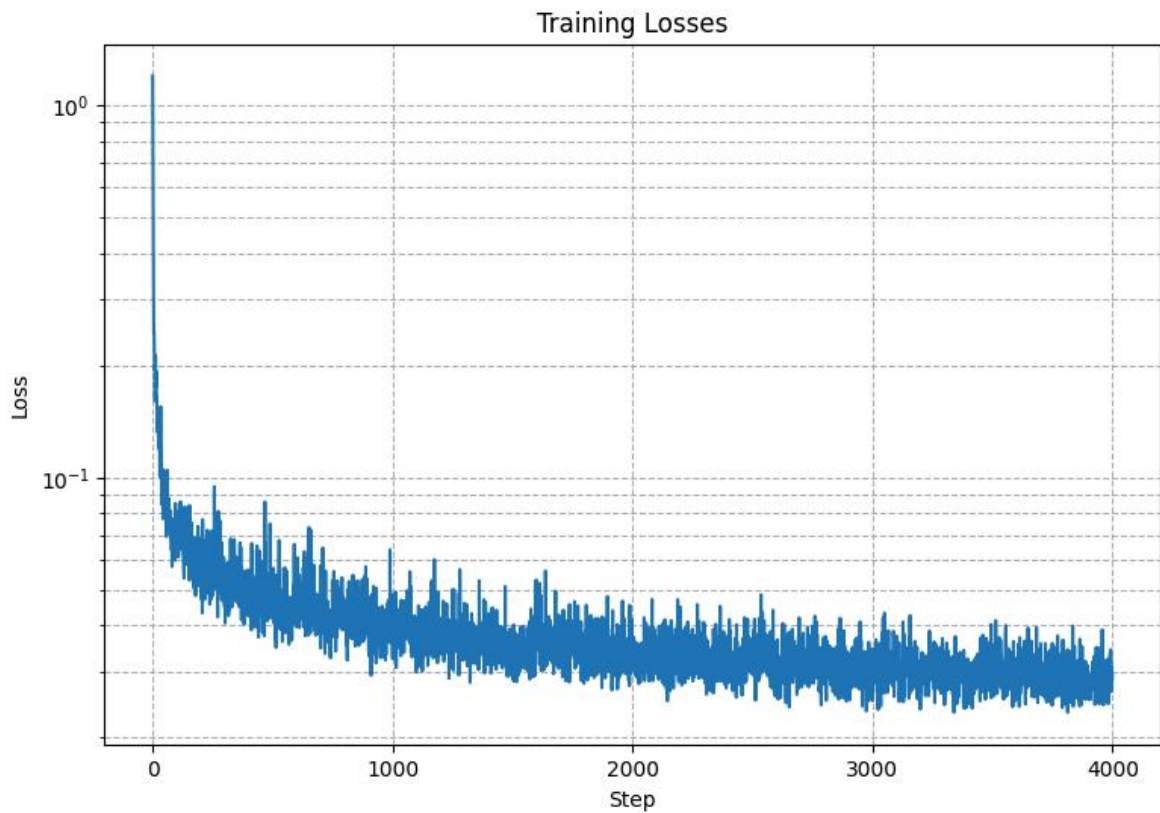
```
In [52]: # Plot training loss curves with respect to first 4000 steps
plt.figure(figsize=(9, 6))
```

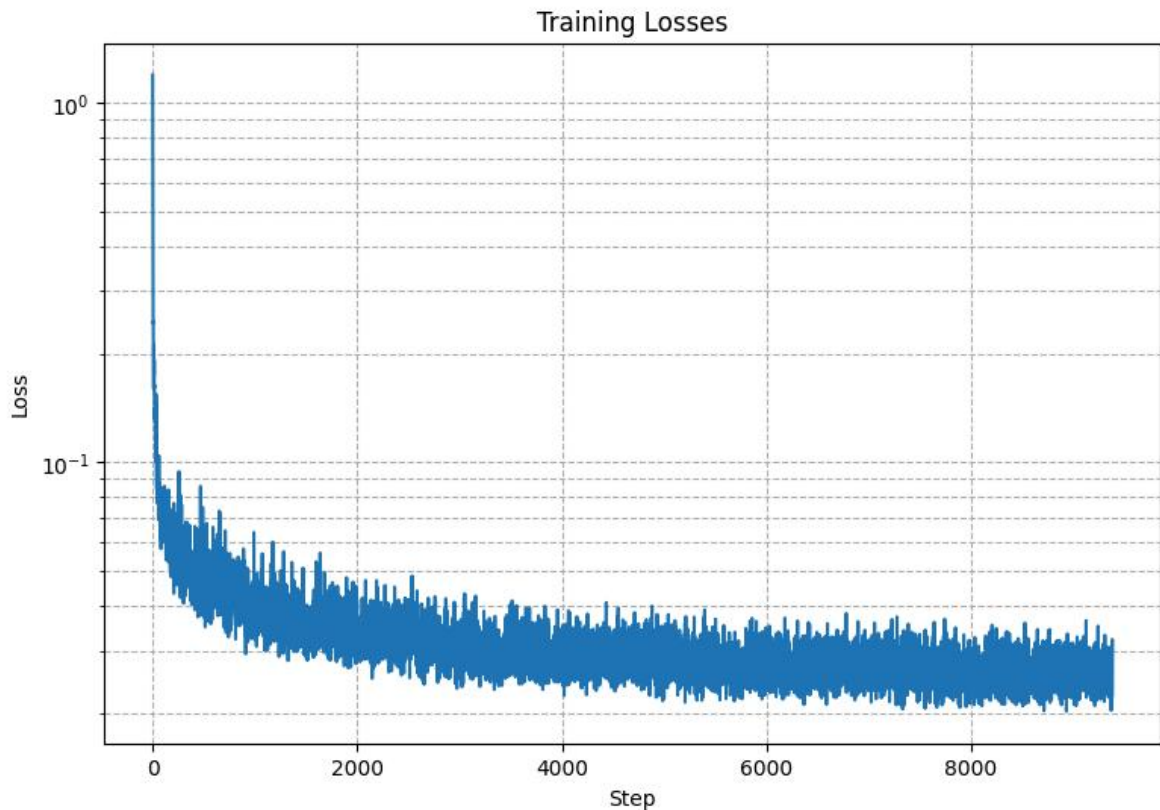
```

plt.semilogy(step_losses[:4000])
plt.xlabel('Step')
plt.ylabel('Loss')
plt.title('Training Losses')
plt.grid(True, which='both', ls='--')
# Only choose 5 ticks(0, 1000, 2000, 3000, 4000) in the x-axis
ax = plt.gca()
ax.xaxis.set_major_locator(ticker.MaxNLocator(nbins=5, integer=True))
plt.show()

# Plot training loss curves with respect to all steps
plt.figure(figsize=(9, 6))
plt.semilogy(step_losses)
plt.xlabel('Step')
plt.ylabel('Loss')
plt.title('Training Losses')
plt.grid(True, which='both', ls='--')
plt.show()

```





Implementing class-conditioned UNet

```
In [53]: class ClassConditionalUNet(nn.Module):
    def __init__(self, in_channels: int, num_classes: int, num_hiddens: int):
        super().__init__()
        self.conv_block1 = ConvBlock(in_channels, num_hiddens)
        self.conv_block2 = ConvBlock(2*num_hiddens, num_hiddens)

        self.down_block1 = DownBlock(num_hiddens, num_hiddens)
        self.down_block2 = DownBlock(num_hiddens, 2*num_hiddens)

        self.flatten = Flatten()

        self.unflatten = Unflatten(2*num_hiddens)

        self.up_block1 = UpBlock(4*num_hiddens, num_hiddens)
        self.up_block2 = UpBlock(2*num_hiddens, num_hiddens)

        # Time-embedding FCBlocks
        self.fcblock1 = FCBlock(1, 2*num_hiddens)
        self.fcblock2 = FCBlock(1, num_hiddens)

        # Class-embedding FCBlocks
        self.fcblock3 = FCBlock(num_classes, 2*num_hiddens)
        self.fcblock4 = FCBlock(num_classes, num_hiddens)

        self.conv = nn.Conv2d(in_channels=num_hiddens, out_channels=1, kernel_si

    def forward(self, x: torch.Tensor, c: torch.Tensor, t: torch.Tensor, mask: t
        """
        Args:
            x: (N, C, H, W) input tensor.
            c: (N,) int64 condition tensor.
```

```

        t: (N,) normalized time tensor.
        mask: (N,) mask tensor. If not None, mask out condition when mask ==

Returns:
    (N, C, H, W) output tensor.
    """
    assert x.shape[-2:] == (28, 28), "Expect input shape to be (28, 28)."
    # First encode c into one-hot vector
    num_classes = self.fcblock3.fc1.in_features
    c_oh = F.one_hot(c, num_classes).float() # (N, 10)
    # Mask out condition when mask == 0
    if mask is None:
        mask = torch.ones_like(c, dtype=c_oh.dtype) # (N, )
    c_oh = c_oh * mask.unsqueeze(-1)

    # en_%d represents intermediate variables in encoder stage
    en1 = self.conv_block1(x)
    en2 = self.down_block1(en1)
    en3 = self.down_block2(en2)

    flat = self.flatten(en3)

    # de_%d represents intermediate variables in decoder stage
    # add_%d represents the time conditioning signal embedded in decoder stage
    # cat_%d represent concatenated intermediate variables from both encoder
    de1 = self.unflatten(flat)
    add1 = self.fcblock1(t.unsqueeze(-1)).reshape(t.shape[0], -1, 1, 1) # (N, 1, 1, 1)
    add3 = self.fcblock3(c_oh).reshape(c_oh.shape[0], -1, 1, 1) # (N, 2D, 1, 1)
    de1 = de1 + add1 + add3
    cat1 = torch.cat([de1, en3], dim=1)

    de2 = self.up_block1(cat1)
    add2 = self.fcblock2(t.unsqueeze(-1)).reshape(t.shape[0], -1, 1, 1) # (N, 1, 1, 1)
    add4 = self.fcblock4(c_oh).reshape(c_oh.shape[0], -1, 1, 1) # (N, D, 1, 1)
    de2 = de2 + add2 + add4
    cat2 = torch.cat([de2, en2], dim=1)

    de3 = self.up_block2(cat2)
    cat3 = torch.cat([de3, en1], dim=1)

    de4 = self.conv_block2(cat3)

    y = self.conv(de4)
    return y

```

```

In [54]: def ddpm_forward(
    unet: ClassConditionalUNet,
    ddpm_schedule: dict,
    x_0: torch.Tensor,
    c: torch.Tensor,
    p_uncond: float,
    num_ts: int,
) -> torch.Tensor:
    """Algorithm 3 of the DDPM paper.

    Args:
        unet: ClassConditionalUNet
        ddpm_schedule: dict
        x_0: (N, C, H, W) input tensor.

```



```

        c: (N,) int64 condition tensor.
        p_uncond: float, probability of unconditioning the condition.
        num_ts: int, number of timesteps.

Returns:
    (,) diffusion loss.
"""
unet.train()
# YOUR CODE HERE.
N = x_0.shape[0]

# Generate time steps randomly, and normalize to [0, 1]
# Note in the algorithm, the random time steps are from 1 to num_ts(T)
t = torch.randint(1, num_ts+1, (N,), device=x_0.device, dtype=torch.long)
t_norm = t / num_ts

# Generate gaussian noise
eps = torch.randn_like(x_0)

# Generate x_t with alpha_bar
# The shape of t is (N,), so the shape of alpha_bar is (N, )
alpha_bar = ddpm_schedule['alpha_bars'][t]
alpha_bar = alpha_bar.reshape(N, 1, 1, 1) # alpha_bar: (N, 1, 1, 1)
x_t = torch.sqrt(alpha_bar)*x_0 + torch.sqrt(1. - alpha_bar)*eps

# Make dropout mask for classifier-free guidance
mask = torch.rand(N, device=x_0.device)
mask = (mask > p_uncond).float()

# Let UNet predict the noise
pred_eps = unet(x_t, c, t_norm, mask)

# Calculate diffusion loss
diffuse_loss = F.mse_loss(pred_eps, eps, reduction="mean")
return diffuse_loss

```

```

In [55]: @torch.inference_mode()
def ddpm_sample(
    unet: ClassConditionalUNet,
    ddpm_schedule: dict,
    c: torch.Tensor,
    img_wh: tuple[int, int],
    num_ts: int,
    guidance_scale: float = 5.0,
    seed: int = 0,
) -> torch.Tensor:
    """Algorithm 4 of the DDPM paper with classifier-free guidance.

    Args:
        unet: ClassConditionalUNet
        ddpm_schedule: dict
        c: (N,) int64 condition tensor. Only for class-conditional
        img_wh: (H, W) output image width and height.
        num_ts: int, number of timesteps.
        guidance_scale: float, CFG scale.
        seed: int, random seed.

    Returns:
        (N, C, H, W) final sample.
        (N, T_animation, C, H, W) caches.

```

```

"""
UNET
"""
UNET.eval()
# YOUR CODE HERE.
H, W = img_wh
N = c.shape[0]
C = 1 # The number of image channels in MINST is 1

# Get device of UNet
UNET_device = next(UNET.parameters()).device

# Get tensors in ddp_schedule
beta = ddp_schedule['betas']
alpha = ddp_schedule['alphas']
alpha_bar = ddp_schedule['alpha_bars']

# Generate x_t(t=T) as pure noise
torch.manual_seed(seed)
x_t = torch.randn(N, C, H, W) # x_T: (N, C, H, W)
x_t = x_t.to(UNET_device)
# Cache list
caches = [x_t.clone()]

# Stepwise backsampling from t=num_ts(T) to t=1
for t in range(num_ts, 0, -1):
    t_norm = torch.tensor([t], device=UNET_device, dtype=torch.float32) / num_ts
    t_norm = t_norm.repeat(N)

    # Classifier-free guidance
    mask_u = torch.zeros(N, device=UNET_device)
    mask_c = torch.ones(N, device=UNET_device)
    pred_eps_u = UNET(x_t, c, t_norm, mask_u)
    pred_eps_c = UNET(x_t, c, t_norm, mask_c)
    pred_eps = pred_eps_u + guidance_scale * (pred_eps_c - pred_eps_u)

    # Get beta_t/alpha_t/alpha_bar_t
    beta_t = beta[t]
    alpha_t = alpha[t]
    alpha_bar_t = alpha_bar[t]

    # Generate noise z
    if t > 1:
        z = torch.randn_like(x_t) # z: (N, C, H, W)
    else:
        z = torch.zeros_like(x_t) # z: (N, C, H, W)

    # According to the formula: x_{t-1} = 1/sqrt(alpha_t) [ x_t - (1-alpha_t)/sqrt(1-alpha_bar_t) * pred_eps ]
    coef_of_pred_eps = (1 - alpha_t) / torch.sqrt(1 - alpha_bar_t)
    x_t = (x_t - coef_of_pred_eps * pred_eps) / torch.sqrt(alpha_t) + torch.sqrt(beta_t) * z

    # Add result into the cache list
    if t % 10 == 0:
        caches.append(x_t)

caches_tensor = torch.stack(caches, dim=0) # (T_animation, N, C, H, W)
caches_tensor = caches_tensor.permute(1, 0, 2, 3, 4) # (N, T_animation, C, H, W)

return x_t, caches_tensor

```

```

In [56]: class DDPM(nn.Module):
def __init__(

```

```

        self,
        unet: ClassConditionalUNet,
        betas: tuple[float, float] = (1e-4, 0.02),
        num_ts: int = 300,
        p_uncond: float = 0.1,
    ):
        super().__init__()
        self.unet = unet
        self.betas = betas
        self.num_ts = num_ts
        self.p_uncond = p_uncond
        self.ddpm_schedule = nn.ParameterDict(ddpm_schedule(betas[0], betas[1]),

def forward(self, x: torch.Tensor, c: torch.Tensor) -> torch.Tensor:
    """
    Args:
        x: (N, C, H, W) input tensor.
        c: (N,) int64 condition tensor.

    Returns:
        (,) diffusion loss.
    """
    return ddpm_forward(self.unet, self.ddpm_schedule, x, c, self.p_uncond,

@torch.inference_mode()
def sample(
    self,
    c: torch.Tensor,
    img_wh: tuple[int, int],
    guidance_scale: float = 5.0,
    seed: int = 0,
):
    return ddpm_sample(self.unet, self.ddpm_schedule, c, img_wh, self.num_ts

```

Training the Class-conditioned UNet

- Plot the loss curve
- Sample results on the test set

```
In [60]: device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(device)
```

cuda

```
In [61]: train_dataset = MNIST(root='data', download=True, transform=ToTensor(), train=True)
train_dataloader = DataLoader(dataset, batch_size=128, shuffle=True)
test_dataset = MNIST(root='data', download=True, transform=ToTensor(), train=False)
test_dataloader = DataLoader(dataset, batch_size=40, shuffle=True)

# TODO
net = ClassConditionalUNet(in_channels=1, num_classes=10, num_hiddens=64)
model = DDPM(net, betas=(0.0001, 0.02), num_ts=300, p_uncond=0.1).to(device)

# Determine loss function, scheduler and optimizer
from torch.optim.lr_scheduler import ExponentialLR

num_epochs = 20
```

```

learning_rate = 1e-3
gamma = 0.1 ** (1/num_epochs)

optimizer = optim.Adam(model.unet.parameters(), lr=learning_rate)
scheduler = ExponentialLR(optimizer, gamma=gamma)
# criterion = nn.MSELoss()

```

```

In [62]: # Record the losses for each batch/step
step_losses=[]
checkpoints = {0, 4, 9, 14, 19}

# Train the diffusion model
for epoch in range(0, num_epochs):
    model.train()

    for x_0, c in train_dataloader:
        x_0 = x_0.to(device)
        c = c.to(device)

        # Forward pass
        # Note that DDPM forward() returns the diffusion loss
        loss = model(x_0, c)

        # Backward pass
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        step_losses.append(loss.item())

    # Update the Learning rate
    scheduler.step()

    # Calculate the average loss over the current epoch
    avg_loss = sum(step_losses[-len(train_dataloader):]) / len(train_dataloader)
    print(f"Epoch {epoch+1:2d} | avg train loss: {avg_loss:.4e}")

    # Visualize on certain epochs
    if epoch in checkpoints:
        model.eval()
        n_rows, n_cols = 4, 10
        num_samples = n_rows * n_cols

        # Construct class labels tensor
        c = torch.arange(n_cols, device=device).repeat(n_rows)

        # Set guidance_scale to be 5.0, and we use 42 as a fixed seed
        samples, caches = model.sample(c, (28, 28), 5.0, seed=42) # samples: (40, 28, 28)

        save_ddpm_gif(caches, f"gif/Class_Unet_epoch_{epoch+1}.gif", n_rows, n_cols)

        big_grid = vutils.make_grid(samples, nrow=n_cols, padding=0, normalize=False)

        plt.figure(figsize=(10, 4))
        plt.imshow(big_grid[0].cpu(), cmap='gray', vmin=0, vmax=1)
        plt.axis('off')
        plt.title(f"Samples at Epoch {epoch+1}")
        plt.show()

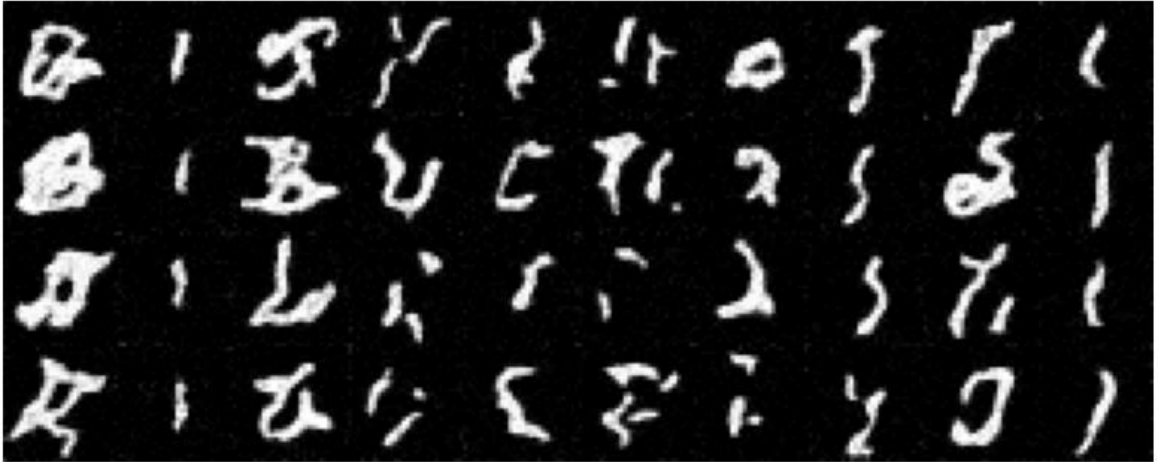
        # grid = samples.squeeze(1).cpu() # (40, 28, 28)

```

```
# fig, axes = plt.subplots(n_rows, n_cols, figsize=(n_cols, n_rows), gri
# for i, ax in enumerate(axes.flat):
#     ax.imshow(grid[i], cmap='gray')
#     ax.axis('off')
# plt.subplots_adjust(left=0, right=1, top=1, bottom=0)
# plt.suptitle(f"Samples at Epoch {epoch+1}")
# plt.show()
```

Epoch 1 | avg train loss: 7.0319e-02

Samples at Epoch 1



Epoch 2 | avg train loss: 4.2343e-02

Epoch 3 | avg train loss: 3.6812e-02

Epoch 4 | avg train loss: 3.3767e-02

Epoch 5 | avg train loss: 3.1603e-02

Samples at Epoch 5



Epoch 6 | avg train loss: 3.0256e-02

Epoch 7 | avg train loss: 2.9326e-02

Epoch 8 | avg train loss: 2.8752e-02

Epoch 9 | avg train loss: 2.8086e-02

Epoch 10 | avg train loss: 2.7834e-02

Samples at Epoch 10



Epoch 11 | avg train loss: 2.7348e-02
Epoch 12 | avg train loss: 2.6963e-02
Epoch 13 | avg train loss: 2.6709e-02
Epoch 14 | avg train loss: 2.6306e-02
Epoch 15 | avg train loss: 2.6284e-02

Samples at Epoch 15



Epoch 16 | avg train loss: 2.5986e-02
Epoch 17 | avg train loss: 2.5821e-02
Epoch 18 | avg train loss: 2.5569e-02
Epoch 19 | avg train loss: 2.5319e-02
Epoch 20 | avg train loss: 2.5377e-02

Samples at Epoch 20



```
In [63]: # Plot training loss curves with respect to first 4000 steps  
plt.figure(figsize=(9, 6))
```

```

plt.semilogy(step_losses[:4000])
plt.xlabel('Step')
plt.ylabel('Loss')
plt.title('Training Losses')
plt.grid(True, which='both', ls='--')
# Only choose 5 ticks(0, 1000, 2000, 3000, 4000) in the x-axis
ax = plt.gca()
ax.xaxis.set_major_locator(ticker.MaxNLocator(nbins=5, integer=True))
plt.show()

# Plot training loss curves with respect to all steps
plt.figure(figsize=(9, 6))
plt.semilogy(step_losses)
plt.xlabel('Step')
plt.ylabel('Loss')
plt.title('Training Losses')
plt.grid(True, which='both', ls='--')
plt.show()

```

